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1)

$P(n) := \exists_l [l \in \mathbb{Z} : 2^{2n} - 1 = 3 \cdot l]$

$\exists$-elim $\rightarrow 0 \in \mathbb{Z}$, we have our witness

(B) $2^{2 \cdot 0} - 1 = 0 = 3 \cdot 0 \rightarrow (\exists^*\text{-intro})\ P(0)$

var $k$; $k \in \mathbb{N}$

$P(k)$

(IH) Pick an $l$ with $l \in \mathbb{Z}$ and $2^{2k} - 1 = 3 \cdot l$

$$\begin{aligned} 2^{2(k+1)} - 1 &= 2^{2k+2} - 1 \\ &= 2^2 \cdot 2^{2k} - 1 \\ &= 4 \cdot (2^{2k} - 1 + 1) - 1 \\ &\overset{IH}{=} 4 \cdot (3 \cdot l + 1) - 1 \\ &= 4 \cdot 3 \cdot l + 4 - 1 \\ &= 4 \cdot 3 \cdot l + 3 \\ &= 4 \cdot 3 (l + 1) \\ &= 3(4l + 4) \end{aligned}$$

Since $(4l + 4) \in \mathbb{Z}$ with $\exists^*$-intro we get $P(k+1)$

$P(k+1)$

(S) $\forall_k [k \in \mathbb{N} : P(k) \Rightarrow P(k+1)]$

{Induction on (B) and (S):}

$\forall_n [n \in \mathbb{N} ; P(n)]$